Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions

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Logged bandit feedback problem

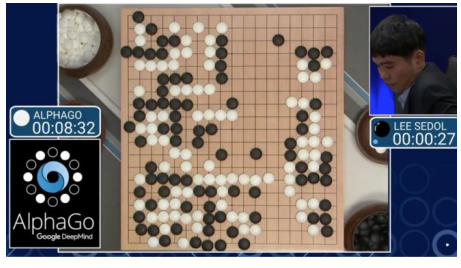
Problem: In interactive systems, full information is not always available.





We wish to learn to **take actions** so as to **maximize a reward** signal...





... while **ensuring safe/affordable learning** for real-world applications with **continuous** treatment (drug dosage, online bidding, etc.).

Counterfactual Risk Minimization[1]

Context $x \sim \mathcal{D}_X$ in $\mathcal{X}$, actions $a \sim \pi(\cdot|x)$ and losses $y \sim D_Y(\cdot|x, a)$

Definition

For any policy $\pi \in \Pi$, and feedback loss y, the expected *risk* is:

$$R(\pi) = \mathbb{E}_{(x,a,y)\sim\mathcal{D}_\pi}[y] = \mathbb{E}_{x\sim \Pr(\mathcal{X})}\, \mathbb{E}_{a\sim\pi_0(a|x)}\left[y\frac{\pi(a|x)}{\pi_0(a|x)}\right]$$

which motivates the ***inverse propensity scoring*** (IPS, Horvitz and Thompson, 1952) estimators:

$$\hat{R}_{\text{IPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \frac{\pi(a_i|x_i)}{\pi_{0,i}}.$$

Goal: find a policy π that minimizes $\hat{R}(\pi)$ with regularizers $\Omega(\pi)$.

$$\hat{\pi} \in \arg\min_{\pi\in\Pi} \hat{R}(\pi) + \Omega(\pi).$$

[1] Swaminathan and Joachims, 2015a.

Contributions

In this work, we list the following contributions:

- **Modeling:** we propose a new *regularized estimator* with a *contextual modelling* for the *continuous action* case.
- **Optimize:** We show the importance of optimization with *soft-clipping* and *proximal point algorithms*.
- **Evaluate**: We release a new large-scale real-world *dataset*, and show how to perform *offline evaluation*.

Modeling: continuous action setup

In this continous[2] setup, we consider **stochastic** parametrized policies, $\theta = (\theta_\mu, \sigma)$.

Baseline contextual modelling of the mean of the pdf μ (σ std))

- **constant**, context-independent parameterization $\mu = \theta_\mu$
- **stratified** piecewise constant: partition of the feature space $\mathcal{X}$
- **linear** $\mu = \theta_\mu^\top x$, **non-linear** $\mu = \theta_\mu^\top \varphi(x)$

Instead, we learn a loss predictor of the form $\eta(x, a) = w^\top \phi(x, a)$

Counterfactual Loss Predictor: we build a policy centered around the prediction $\mu_\gamma(x) = \text{argsoftmax}_{a \in \tilde{\mathcal{A}}} \eta(x, a)$ (γ, σ hyperparameter)

$$\hat{\pi}_{\text{CLP}}(\cdot|x) = \mathcal{N}(\mu_\gamma(x), \sigma^2)$$

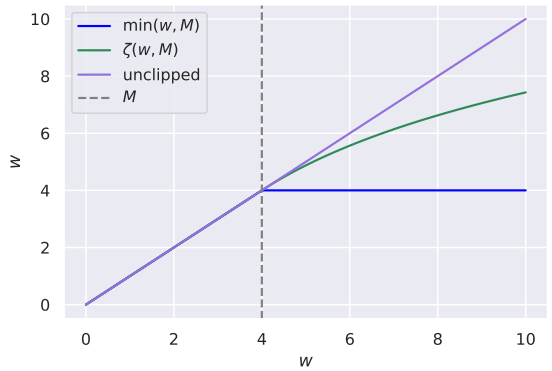
[2] Kallus and Zhou, 2018.

Modeling: a regularization for gradient-based optimization

Counterfactual estimators suffer from **unbounded variance**[3].

Clipping (Bottou et al., 2013) the importance sampling weights:

$$\hat{R}^M_{\text{cIPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \min\left\{\frac{\pi(a_i|x_i)}{\pi_{0,i}}, M\right\}.$$



$$\zeta(w_i, M) = \begin{cases} w_i \\ \alpha(M)\log\left(w_i + \alpha(M) - M\right) \end{cases}$$

where $\alpha(M)\log(\alpha(M)) = M$

Soft-clipping continuous and differentiable operator (scIPS):

$$\hat{R}^M_{\text{scIPS}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} y_i \zeta\left(\frac{\pi(a_i|x_i)}{\pi_{0,i}}, M\right)$$

[3] Bottou et al., 2013.

Optimization: proximal point algorithms

The **learning problem** is non convex (clipping, self-normalized[a], variance penalty, etc.[b]):

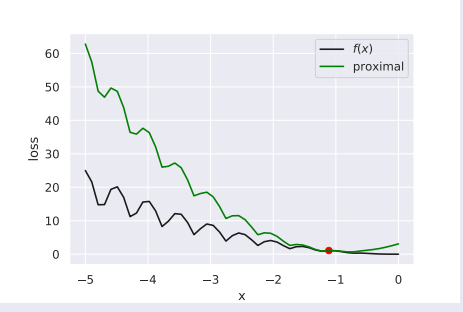
$$\hat{\theta} \in \arg\min_{\theta \in \Theta} \hat{R}(\pi_\theta) + \Omega(\pi_\theta).$$

[a] Swaminathan and Joachims, 2015b.
[b] Dudık, Langford, and Li, 2011.

Proximal Point Method (Rockafellar, 1976)

Approximately solve a sequence of subproblems that are better conditioned and easier to solve than $\mathcal{L}$

$$\theta_k \approx \arg\min_{\theta} \left(\mathcal{L}(\theta) + \frac{\kappa}{2} \|\theta - \theta_{k-1}\|_2^2 \right)$$



Evaluation: potential prediction setup

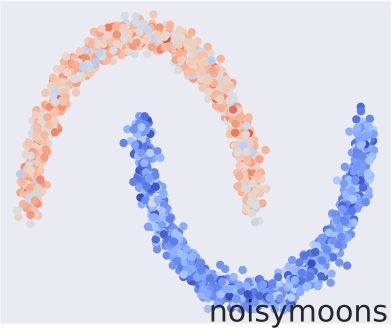
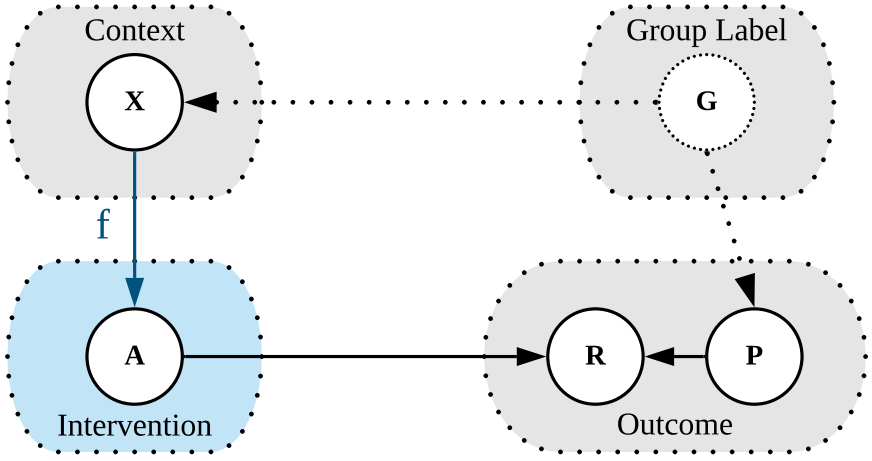


Figure: Example of synthetic 2D features with potentials

Actions $a \sim A$ should attempt predict what is the potential p thus inducing a **reward R**.

Potential prediction setup

Features $x \sim X$ belongs to a **class/group** $g \sim G$ and have **potentials** $p \sim P \mid (G = g)$.



Evaluation: releasing a new real-world dataset

Causal advertising setup

For user **features** $x \sim X$ an **action** $a \sim A$ is taken to decide which amount to **bid** B. A **cost** $c \sim C$ and a number of **sales** S define a **reward** R.

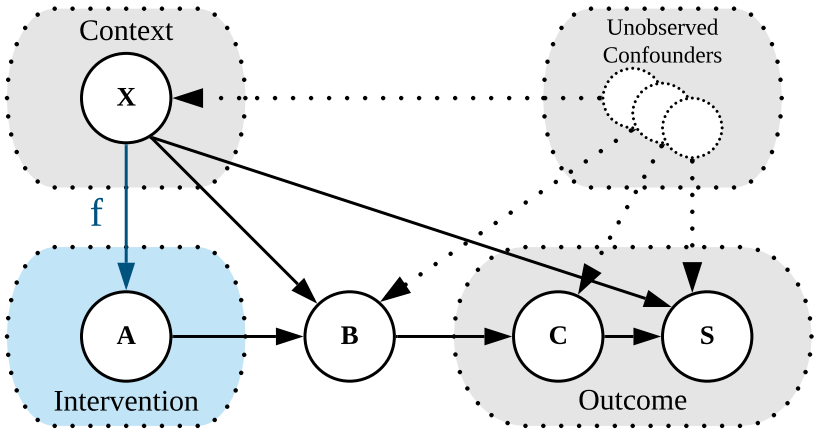


Table: Summary statistics - Criteo Off Policy Continuous Action Dataset

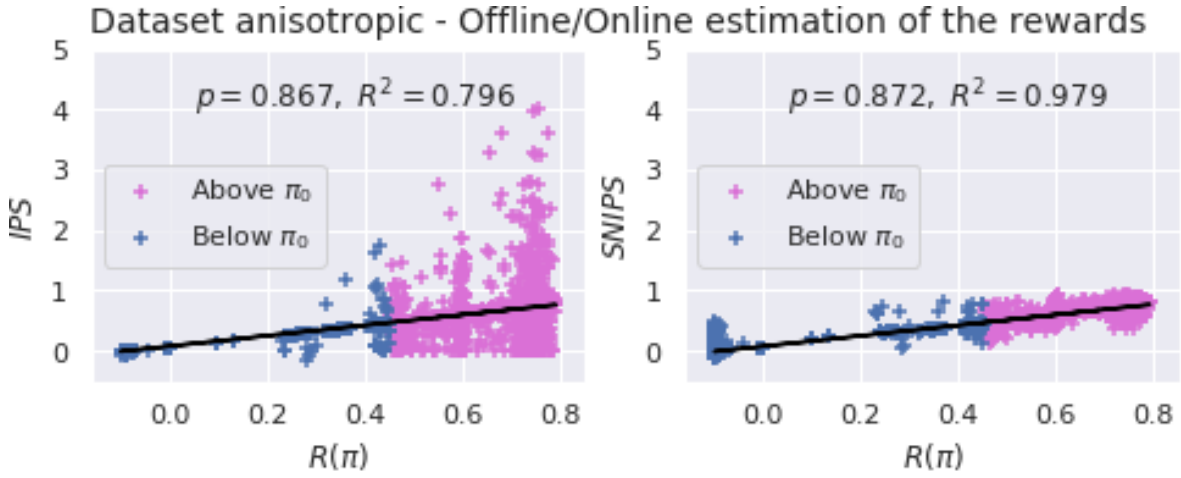
N	d	$\mathbb{E}[R]$	$\mathbb{V}[R]$	$\mathbb{E}[A]$	$\mathbb{V}[A]$
119,971,451	3	11.37	9455.01	1.00	0.01

Evaluation: offline protocol

When no access to online evaluation with the reward function:

- What offline metric should we use?
- How to know that the solution is statistically significant?

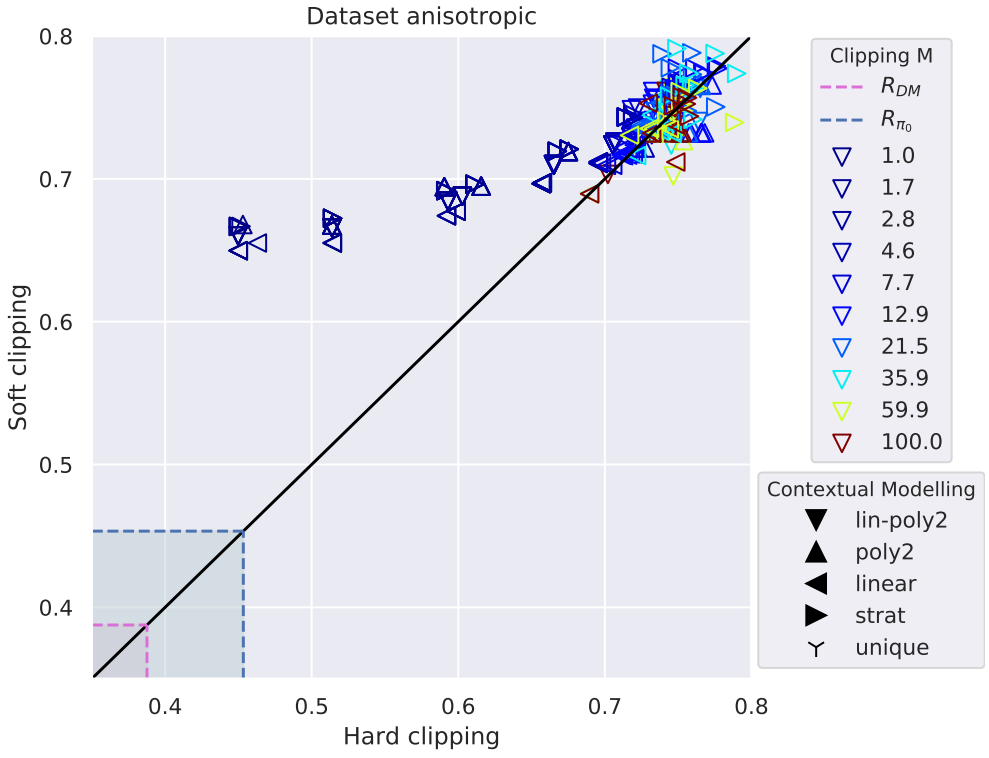
Offline metric: self-normalized importance sampling.



Statistical significance: discard solutions with effective sample sizes (Owen, 2013) which are low (see criteria in paper).

Evaluation: scIPS

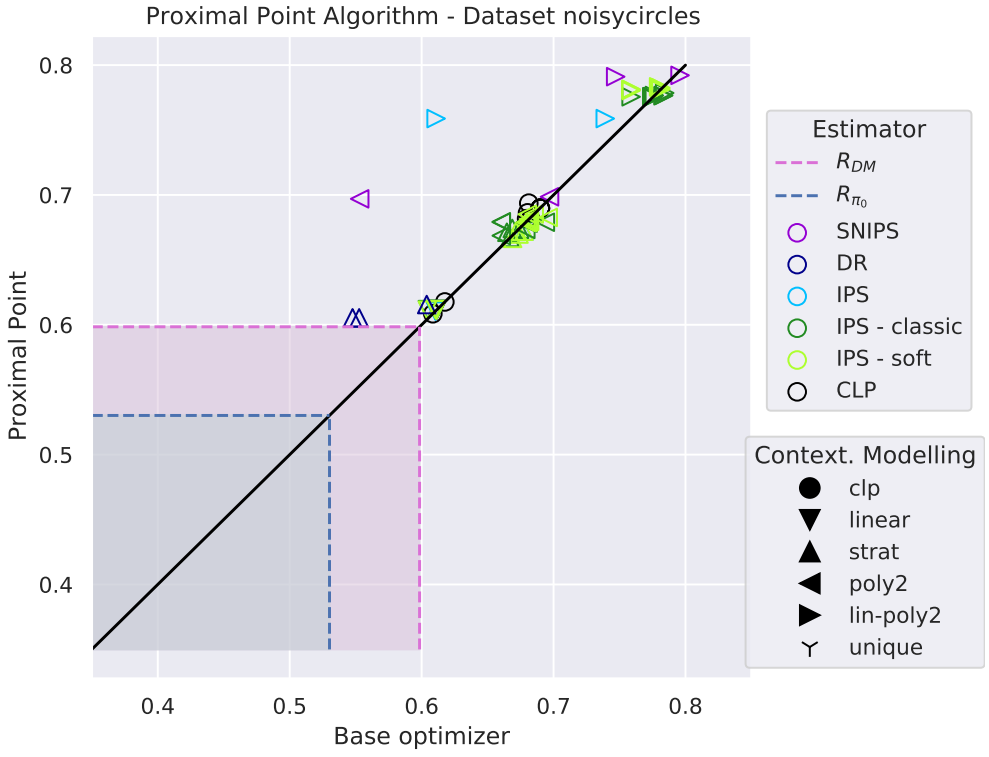
We compare test rewards for soft- vs hard-clipping and provide confidence level metrics with success ratio ρ.



Dataset	Anis.	Criteo
ρ^{Soft}	**0.713**	**0.500**
ρ^{Hard}	0.216	0.250
ρ^{Soft}_{68}	**0.673**	**0.250**
ρ^{Hard}_{68}	0.215	0.083

Evaluation: proximal point algorithm

We compare test rewards for proximal-point vs the simpler gradient-based method and provide confidence level metrics with success ratio ρ.



Dataset	N.Circles	Criteo
ρ^{PROX}	**0.722**	**0.881**
ρ^{NONPROX}	0.290	0.644
ρ^{PROX}_{68}	**0.692**	**0.191**
$\rho^{\text{NONPROX}}_{68}$	0.267	0.059

Evaluation: contextual modelling and other

Table: Comparison of evaluation metrics according to contextual parameterizations

	NoisyCircles	NoisyMoons	Anisotropic	Criteo
CLP	**0.697** $\pm$ 0.07	0.733 $\pm$ 0.07	**0.788** $\pm$ 0.06	**11.40** $\pm$ 0.16
Strat	0.670 $\pm$ 0.00	**0.748** $\pm$ 0.00	0.739 $\pm$ 0.02	2.20 $\pm$ 1.36
Linear	0.599 $\pm$ 0.00	0.725 $\pm$ 0.00	0.633 $\pm$ 0.00	9.45 $\pm$ 0.77
Poly2	0.611 $\pm$ 0.00	0.723 $\pm$ 0.00	0.627 $\pm$ 0.00	8.78 $\pm$ 0.51
Lin-poly2	0.530 $\pm$ 0.00	0.624 $\pm$ 0.00	0.658 $\pm$ 0.00	7.47 $\pm$ 0.41

Table: SNIPS estimates of the test reward on the Criteo dataset

	Hard-clipping	Soft-clipping
Non-Prox	11.30 $\pm$ 0.06	11.28 $\pm$ 0.09
Prox	11.22 $\pm$ 0.09	**11.37** $\pm$ 0.08

Take-home message and future work

Take-home message:

- **Optimization** is equally important as **modeling** for the resulting non-convex cost functions. To that effect, we introduce smooth clipping, and use proximal point algorithms.
- **Evaluation** for real-world offline datasets has received little attention in continuous settings: we provide a real-world dataset and a protocol for performance measurement.

Future research:

- Succession of offline and online learning with deployment of policies in production systems.

References

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